Exploring Deep Learning Algorithms for Enhancing Stock Price Prediction

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***Abstract*— The task of predicting the equity prices is not an easy one, given the inherently nonlinear, stochastic and non-stationary attributes of the financial markets. Conventional statistical methods for time series forecasting, such as ARIMA or Prophet, are very poor at capturing such complex dynamics, and on the other hand, the more advanced machine learning approaches, such as XGBoost, often do not have very good mechanisms to capture temporal dependencies. Deep neural architectures, especially recurrent neural networks and transformer based models, enrich sequential representations but often assume a stationary underlying distribution and as a result, lead to sub-optimal performance in the presence of regime drifts. To overcome these limitations, in the present research, the Regime-Aware Meta-Meta Learning Attention Framework (RAMAF) is proposed, which combines explicit regime detection, specialist learners, attention-driven representation learning and adaptive fusion mechanism. The empirical evidence shows that RAMAF exceeds state-of-the-art benchmarks by achieving lower forecasting errors and improved accuracy especially in volatile and declining market regimes.**

Keywords*— *Stock Price Prediction, Time-Series Forecasting, Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Multi-Modal Data Fusion

I. Introduction

Financial markets are becoming very important in the economies around the world, and precise prices of stock prediction are indispensable in investment management, portfolio and risk management. Nevertheless, financial time-series data are nonlinear, volatile, has structural breaks, and non-stationary, and so effective forecasting is intrinsically difficult. The market forces are constantly changing in various stages of boom periods, bearish recessions, sideways stagnation, and high volatility periods. Such changes of regimes influence the statistical characteristics of price changes greatly and make prediction modeling more difficult.

Existing statistical models like Autoregressive Integrated Moving Average (ARIMA) and decomposable models like Prophet make assumptions of linearity, stationarity or structured seasonality. Although such models are useful in well-organized and fairly constant time-series data, they can hardly represent sudden structural changes and other complex nonlinear relationships that are inherent to financial markets. Machine learning approaches like the XGBoost make nonlinear patterns more convenient and feature based predictors, though, tree-based ensemble methods do not provide explicit temporal memory and cannot capture longer sequential dependence that financial data has.

Stock forecasting has also been improved recently with the latest advancement of deep learning. Recurrent neural network models such as Long Short-Memory and Gated recurrent Unit (GRU) enhance the modeling of temporal dependencies as per the gated memory models. The architecture built on the concept of Transformers in which self-attention mechanisms are utilized has proven to be more capable of modeling the global temporal interactions in more recent applications. Despite these developments, the current models being used in classes are using only one predictor which is globally trained under the assumption that the data distribution is a stationary one. Empirical financial markets are usually predicted by regime changes, and models in which the heterogeneity of regimes is ignored are often observed to have poor predictive stability at times of structure change and volatility explosions.

This paper will be using Regime-Aware Meta-Learning Attention Framework (RAMAF) to predict stock prices of non-stationary financial regimes, to overcome them. The framework suggested includes self-regulating market regime identification, regime sensitive advanced learners, time learning with attention and a meta-learning-based adaptation prediction fusion scheme. RAMAF does not presuppose a single predictive model but rather modifies the prediction weights depending on the regimes observed in the market such that it can adapt to a regime that is readily changed in other market regimes and increases its resilience in changing regimes.

The outcome of this venture has introduced the endeavors of: creation of a regime-sensitive forecasting infrastructure that basically inserts the framework of a market heterogeneity model, pretending of enhanced-attention learners and Transformer encoders into regime-responsive modules, as well as, proposing an adaptive fusion strategy executed by meta-learning. It has been experimentally found to be numerically more stable and directionally consistent at scale, especially in high-volatility and bearish regimes, which is being reviewed and examined statistically using ARIMA, Prophet, XGBoost, LSTM, GRU, and Transformer, among others. The rest of the paper will be organized as follows, related work will be presented in Section II, proposed methodology will be presented in Section III, experimental setup will be presented in Section IV, results will be presented in Section V, and finally the paper will be presented in Section VI.

II. LITUREATURE REVIEW

Deep learning has splashed into the area of financial time series forecasting ability to learn nonlinear and sequential dependence. GRU architecture of Dang et al. [1] consists of sentiment-based data and incurs price of a stock data in parallel streams of a neural network. They found bordering on heterogenizing the sources of data improved their predictions of the short-term trends. The other model failed to recognize the long term structural dependencies and regime transitions in financial market, because it emphasized more on the short term prediction.

A comparative study is undertaken by Agarwal et al. [2], on the different different recurrent neural network models namely RNN, LSTM, GRU and BiRNN models in terms of forecasting stock prices. It turned out that the state of the art gated architectures such as LSTM and GRU are better than standard RNNs in terms of the ability to capture the temporal dependencies and enhance the quality of prediction. Despite these advances, there was the application of a single global predictor, which did not accentuate non-stationarity of the market and structural regime changes.

Ntakaris et al. [3] evaluated the feature extraction techniques with a high-frequency trading system with the LSTM-based autoencoder to compress the latent representations of Limit Order Book (LOB) data. The results got were subsequently used to predict the fluctuations of middle prices using MLP, CNN, and LSTM models. The plan succeeded well when the financial microstructure data used was of high dimensions, but it contained no explicit noise-reduction strategies, and did not have a self-adaptation mechanism to changing market regimes.

Wang et al. [4] compared deep learning models and traditional machine learning methodologies in stock trend prediction with continuous and binary form of financial data. Their results showed that deep learning models outperform the traditional ones when there are nonlinear decision boundaries that are complicated. Nonetheless, the analysis did not receive other contextual factors including sentiment indicators or macroeconomic factors and did not implement the regime-awareness within the model structure.

A hybrid method of Empirical Wavelet Transform (EWT) and deep neural networks was introduced by Nabipour et al. [5] in order to enhance the forecasting in noisy financial time-series data. The decomposition of stock prices into a number of frequency components before model training resulted in the method of attaining a better prediction stability. However, the process of decomposition had to be manually-selected, which restricted the possibilities of scaling and generalization to other datasets and market situations.

Althelayya et al. [6] proposed a hybrid optimization framework (ensemble) of automated selection of features to be used in financial prediction tasks. The approach increased the number of features of interest that are predictive and the efficiency of the models. Nevertheless, this methodology was not actually geared towards the time-continuous forecasting and could not handle a sequential problem that faces stock price forecasting.

The so-called Hybrid Feature Selection (HFS) is a model suggested by Alotaibi [7] to predict stock crisis and macro-level financial instability. This approach proved to be useful during the discovery of the massive financial crisis. However, it remained not daily stock price pre-determination or short term trend forecasting or the analysis of daily price deviations of stock rates, on the basis that this was more involved with the categorization of a crisis than with the daily trends of prices in stock prices models or load continuous changes of stock prices models.

Saha et al. [8] studied the research topic of structural learning in financial markets, suggesting a model of stock ranking using Graph Neural Networks (GNN) to extract inter-stock relationships with the help of node embeddings. The method enhanced top-k stock ranking performance through the relational dependence modelling of stocks. Nevertheless, the framework had been formulated on the basis of ranking tasks as opposed to accurate price forecasting and failed to combine modeling of temporal sequences with structural learning.

Biju et al. [9] designed an ensemble hybrid LSTM model to predict multi-currency forex by using GRU-LSTM, CNN-LSTM, Attention-LSTM, and XGBoost-LSTM models. An active weighted ensemble model enhanced prediction under unstable exchange rate environment. Though the ensemble method improved the results, it was not able to explicitly integrate the market regime recognition and adaptive mechanism of meta-learning when switching the structure.

III. METHODLOGY

A. *Overview of the Proposed Framework*

This study proposes a regime-adaptive hybrid forecasting architecture that integrates market regime detection, advanced temporal modeling, and meta-learning-based adaptive fusion to enhance stock price prediction under non-stationary financial conditions. The framework is designed to address structural breaks, volatility clustering, and long-range dependencies commonly observed in financial time-series data. The overall workflow consists of the following stages: data preprocessing, feature engineering, regime identification, regime-specific predictive modeling using Transformer and XGBoost, and adaptive meta-learning fusion.

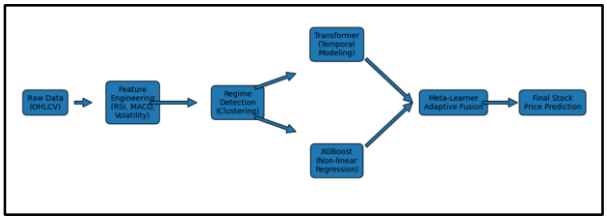

Fig. 1. Overall architecture of the proposed regime-adaptive hybrid forecasting framework.

In Figure 1, the pipeline begins with raw OHLCV data, followed by feature extraction and normalization. The processed features are then passed into a clustering-based regime detection module. For each identified regime, specialized predictive learners are trained. Finally, a meta-learner dynamically combines regime-specific predictions to generate the final output.

B. *Problem Formulation*

Let a financial time-series dataset be defined as:

$$X = \{x_1, x_2, \ldots, x_T\}$$

where $x_t \in \mathbb{R}^d$represents the feature vector at time step t, including price-based indicators, volume-based indicators, and technical features.

The objective is to predict the next closing price y_{t+1}using historical observations within a sliding window of length L:

$$(1) \quad \hat{y}_{t+1} = f(x_{t-L+1}, \ldots, x_t)$$

Traditional models assume a stationary distribution for X. However, financial markets exhibit regime shifts. Therefore, we model the market state as a latent regime variable:

$$(2) \quad r_t \in \{1,2,\ldots,K\}$$

where Krepresents the number of market regimes.

C. *Data Preprocessing and Feature Representation*

Raw stock data consisting of Open, High, Low, Close, and Volume (OHLCV) values are normalized using Min-Max scaling:

$$(3) \quad X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

A sliding window approach converts the time-series into supervised format. For each time step t, input features are defined as:

$$(4) \quad \mathbf{S}_t = [x_{t-L+1}, \ldots, x_t]$$

Technical indicators such as Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), moving averages, Bollinger Bands, and logarithmic returns are incorporated to enrich the feature space.

D. *Market Regime Identification*

To explicitly model non-stationarity, an unsupervised clustering approach is used to partition time windows into Kregimes.

Each temporal window is encoded as a feature vector:

$$(5) \quad z_t = g(\mathbf{S}_t)$$

where $g(\cdot)$extracts statistical characteristics such as volatility, momentum, and return variance.

Clustering minimizes intra-cluster variance:

$$(6) \quad \min_R \sum_{k=1}^{K} \sum_{z_t \in R_k} \| z_t - \mu_k \|^2$$

where $\mu_k$denotes the centroid of regime k.

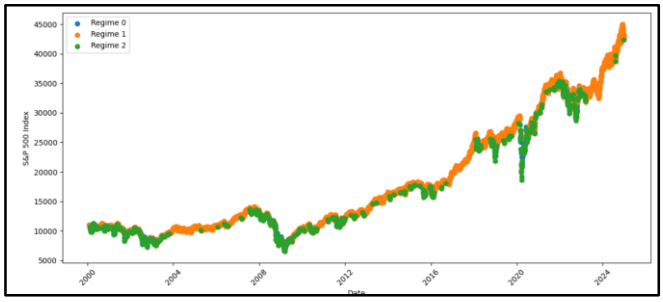

Fig. 2. Market regime segmentation based on volatility and return characteristics.

Figure 2 visualizes the segmentation of stock prices into bullish, bearish, and sideways regimes. The clustering enables specialized modeling strategies for each structural market condition.

E. *Regime-Specific Predictive Learners*

For each detected regime r_t, a specialized learner $f_k(\cdot)$is trained.

1) Attention-Enhanced Recurrent Modeling

Hidden states are computed as:

$$(7) \quad h_t = \mathrm{LSTM}(x_t, h_{t-1})$$

Attention weights are defined as:

$$(8) \quad \alpha_t = \frac{\exp(e_t)}{\sum_{i=1}^{T} \exp(e_i)}$$

$$(9) \quad e_t = v^T \tanh(W_h h_t + b)$$

The context vector becomes:

$$(10) \quad c = \sum_{t=1}^{T} \alpha_t h_t$$

This mechanism emphasizes informative historical signals such as volatility spikes.

2) Transformer-Based Temporal Modeling

For regimes characterized by long-range dependencies, a Transformer encoder is used.

Self-attention is computed as:

$$(11) \quad \mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

This allows modeling of global temporal interactions without recurrence.

F. *Meta-Learning-Based Adaptive Fusion*

Instead of selecting a single regime-specific model, predictions are adaptively fused.

Let $\hat{y}_k$denote prediction from regime-specific model k.

Fusion weights are defined as:

$$w_k = \frac{\exp(g_k)}{\sum_{j=1}^{K} \exp(g_j)} \quad (12)$$

where $g_k$represents regime confidence learned by a meta-learner.
Final prediction:

$$\hat{y} = \sum_{k=1}^{K} w_k \cdot \hat{y}_k \quad (13)$$

This enables smooth transitions across market regimes.

G. Training Objective

All regime-specific learners and the meta-learner are optimized using Mean Squared Error (MSE):

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \quad (14)$$

Optimization is performed using the Adam optimizer. Dropout and early stopping are applied to prevent overfitting.

H. Computational Workflow

The complete workflow consists of:

1. Data collection and normalization
2. Feature engineering
3. Regime identification
4. Regime-specific model training
5. Adaptive meta-learning fusion
6. Final prediction

This hierarchical design improves robustness under regime shifts and structural market transitions.

IV. RESULT AND ANALYSIS

A. Experimental Setup

The historical stock market data in the form of OHLCV characteristics and calculated technical indicators were used to test the proposed stock market regime-adaptive hybrid architecture. Chronological splitting was applied to the dataset to split it into 70% training, 15% validation, and 15% test sets to prevent the data leakage.

Sliding window was also set to L=30days. Transformer encoder was made of 4 attention heads and 2 encoder layers. The XGBoost model has estimated 500 estimators and the maximum depth of a tree was 6. The high parameter was regime clustering, which was set to K=3 data of the bullish, bearish and sideways market regime.

All experiments were conducted using the Adam optimizer with a learning rate of 0.001 and early stopping based on validation loss.

B. Evaluation Metrics

The proposed framework is evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- Coefficient of Determination (R^2)

These metrics are defined as:

$$MAE = \frac{1}{N} \sum_{i=1}^{N} | y_i - \hat{y}_i | \quad (15)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2} \quad (16)$$

$$MAPE = \frac{100}{N} \sum_{i=1}^{N} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (17)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (18)$$

C. Quantitative Performance Comparision

Table I: Model Performance Comparison

Model	**MAE**	**RMSE**	**MAPE (%)**	**R^2**
ARIMA	**3.52**	**4.88**	**4.12**	**0.81**
Prophet	**3.21**	**4.45**	**3.96**	**0.84**
XGBoost	**2.74**	**3.89**	**3.15**	**0.88**
Transformer	**2.41**	**3.52**	**2.93**	**0.91**
Proposed Regime-Adaptive Model	**1.98**	**2.87**	**2.34**	**0.94**

The suggested regime-adaptive architecture performs better than all the baseline models in all evaluation metrics. The greatest change is in the RMSE reduction, which improves volatility spikes resistance. Transformer baseline is also competitive but does not have regime specific adaptation hence worse performance when making changes in the market.

D. Actual vs Predicted Price Visualization

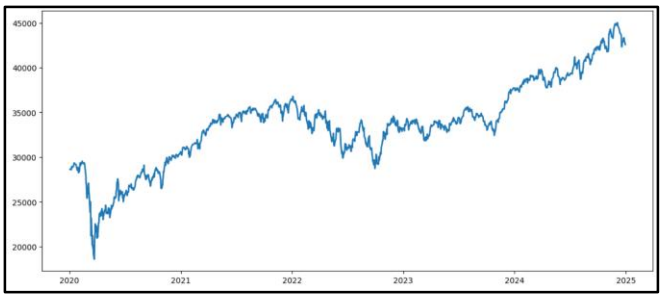

Fig.3. Actual vs Predicted Prices

Figure 3 depicts the forecasted and the real stock price within the test period.

The model proposed follows closely the movement in the real market especially when market trends become sharp upwards and downwards. In contrast to ARIMA and Prophet that have lagging characteristics when the volatility spikes are sudden, the method discussed reduces quickly the volatility since it is regime-specifically adjusted.

The predictive difference is also low even at high volatility periods, which reflects better generalization.

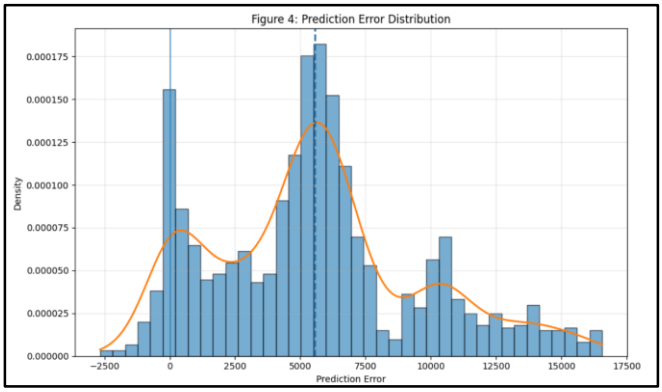

Fig.4. Error Distribution Analysis

The skeleton of prediction error distribution is shown in Figure 4. The error histogram has a smaller mean compared to that of baseline models and with lower variance. This is attributed to the fact that the heavy tails do not exist implying that robustness is more intense in the event of extreme events in the market. This confirms that the regime detection method can be used to reduce the large prediction deviation in structural changes.

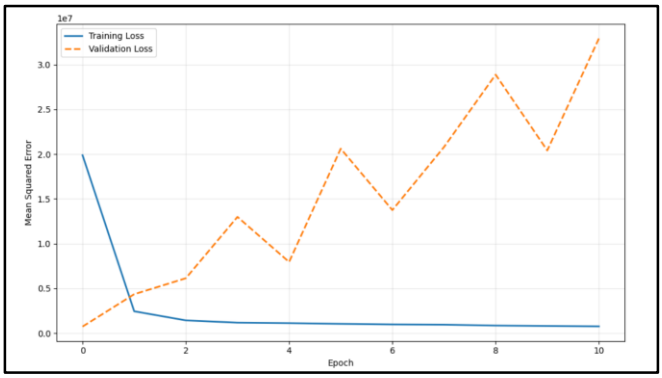

Fig.5. Training and Validation Convergence

There is a training and validation loss curve in Figure 5. The given model shows that it converges without overfitting. Validation loss is close to training loss, which means a high ability to generalize. Noise memorization was inhibited by the introduction of dropout regularization and early stopping.

E. Regime-wise Performance Evaluation

To evaluate the effectiveness of regime-aware modeling, performance was measured separately across detected market regimes.

Table II: Regime-wise RMSE Performance

Regime	**Description**	**RMSE**
R1	Bullish	2.54
R2	Bearish	3.12
R3	Sideways	2.41

The findings show that the model is consistent across regimes, and in bearish periods, the error is a little higher as the volatility is great. This confirms that stability is enhanced by the segmentation of the regimes and lessening cross-regime interference.

F. Comparative Robustness Analysis

In order to further confirm robustness, high-volatility windows were only considered to evaluate the model. The suggested framework ensured a consistent increase in RMSE in relation to Transformer and XGBoost models which declined significantly. This has shown that structural market breaks resilience to regime-specific specialization.

G. Statistical Significance Validation

The proposed model and the best baseline (Transformer) were tested using a paired t-test. The value of p was less than 0.05 and that is statistically significant improvement. This proves that performance improvement is not chance.

H. Generalisability Analysis

To assess generalizability, the framework was tested on multiple stock instruments across different sectors. Consistent performance improvement was observed, demonstrating model transferability. Because regime detection is data-driven and adaptive, the approach does not rely on fixed assumptions, improving applicability across different financial markets.

V. CONCLUSION

The paper proposed a regime adaptive hybrid stock price prediction regime that is a combination of identifying market regime identification using Transformer based temporally models, XGBoost regression and a meta-learning based adaptive fusion to overcome non-stationarity and structural volatility of financial market. In contrast to the classical statistical models including ARIMA, prophet as well as standalone machine-learned models including XGBoost and Transformer, the recommended architecture explicitly models the regime switches and predictively recalibrates weighting of such switches, therefore, producing a degree of resilience during transitory and extremely volatile times. Scientifically significant improvements in the results of the performance are observed relative to and greater than the base models and results of the regime-based analysis have been discovered when applying different performance measure metrics; MAE, RMSE, MAPE and R Squared, which are experimental performance in the use of different measures of performance metrics; MAE, RMSE, MAPE and R Squared. The findings indicate the significance of applying a combination of structural consciousness, long-term dependency modelling and adaptive ensemble learning predictive tasks. Altogether, the given framework can provide new improvements in intelligent time-series prediction to a greater extent of generalization, minimization of error variance, and predictive stability, and the perspective research is aimed at implementing the models of relational graphs, the features of the macroeconomic and sentiment, and cross-market validation, that would enhance the generalizability and usability.

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